

BOUGOUNI, Mali

WMO: 612960

Lat: 11.417N

Lon: 7.500W

Elev: 351

StdP: 97.18

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	16.6	17.7	-0.4	3.8	26.2	1.3	4.3	25.6	4.5	25.6	4.2	26.7	0.5	N/A	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.8	39.8	23.1	39.0	23.0	38.1	23.0	27.8	33.7	27.4	33.1	27.0	32.7	1.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	22.9	31.2	25.9	22.2	30.6	25.5	21.7	30.1	91.3	33.9	89.1	33.4	87.2	32.6	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.9	3.3	2.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	25.9	28.8	31.7	32.2	30.5	28.6	26.7	26.3	26.8	28.1	27.7	25.8		
	DBStd	2.73	1.85	2.26	1.88	1.73	2.09	1.91	1.44	1.28	1.20	1.33	1.35	1.79		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6661	492	526	674	667	637	557	517	506	504	560	531	491		
	CDD18.3	3619	233	292	416	417	378	307	259	247	254	302	281	233		
	CDH23.3	42503	2531	3673	6038	6233	5086	3606	2366	2004	2173	3131	3142	2521		
	CDH26.7	22279	1305	2111	3792	3929	2873	1705	812	584	769	1405	1695	1298		
(14) Wind	WSAvg	0.8	0.9	1.1	1.0	1.1	1.0	0.8	0.5	0.5	0.5	0.5	0.6	1.0		
(15) Precipitation	PrecAvg	1132	1	1	7	42	103	152	260	274	211	73	8	0		
	PrecMax	1488	12	12	43	173	291	255	406	593	392	226	48	20		
	PrecMin	843	0	0	0	0	7	68	141	120	98	3	0	0		
	PrecStd	172	3	2	10	39	56	51	69	84	75	54	13	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.5	39.5	40.9	41.0	40.0	37.5	33.6	32.6	33.5	35.9	37.1	36.7		
		MCWB	20.0	24.2	21.9	23.0	25.2	26.0	26.3	27.0	27.3	23.8	21.8	20.2		
	2%	DB	35.2	38.2	39.7	40.0	38.3	36.1	32.5	31.6	32.5	34.7	36.0	35.4		
		MCWB	19.2	21.8	21.8	23.6	25.7	26.4	26.3	26.5	27.0	25.3	21.7	19.8		
	5%	DB	34.1	37.0	38.9	39.1	37.1	34.6	31.5	30.7	31.6	33.7	35.1	34.2		
		MCWB	18.7	20.9	21.9	23.5	25.9	26.4	26.1	26.2	26.7	25.8	21.8	19.3		
	10%	DB	33.0	35.8	38.1	38.1	35.9	33.4	30.6	29.8	30.6	32.7	34.2	32.9		
		MCWB	18.2	20.2	22.0	23.7	25.9	26.3	25.8	25.8	26.3	25.9	22.0	18.7		
	0.4%	WB	22.2	27.7	28.4	27.2	28.4	28.2	27.3	27.6	28.3	27.7	26.1	23.4		
		MCDB	33.0	38.4	35.2	36.8	35.6	34.0	31.7	31.4	32.3	32.7	33.0	32.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.0	24.8	25.5	26.5	27.7	27.5	26.7	26.9	27.4	27.1	25.2	21.9		
		MCDB	31.9	35.1	35.1	36.1	34.8	33.4	31.1	30.8	31.5	32.2	31.8	32.5		
	5%	WB	19.9	23.3	24.7	26.0	27.2	27.0	26.3	26.4	26.9	26.7	24.6	20.6		
		MCDB	31.9	33.2	35.3	35.5	34.2	32.9	30.6	30.0	31.0	31.7	31.0	31.9		
	10%	WB	18.8	21.9	24.0	25.6	26.7	26.6	26.0	26.0	26.4	26.2	23.8	19.5		
		MCDB	31.5	31.7	34.7	34.6	33.6	32.1	30.1	29.3	30.1	31.1	30.1	31.1		
	5% DB	MDBR	15.2	14.3	13.4	11.8	10.6	9.2	7.4	6.8	8.1	9.9	14.5	15.6		
		MCDBR	16.3	15.5	14.3	13.3	12.1	10.9	8.8	8.3	9.4	11.3	15.5	16.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.7	6.5	5.4	4.0	3.8	3.9	3.8	3.9	4.8	4.1	4.9	6.2		
		MCDBR	15.3	13.4	12.0	11.4	10.9	9.9	8.4	8.0	9.2	10.0	11.9	16.1		
	5% WB	MCWBR	7.0	6.9	5.1	3.8	4.0	4.3	4.1	4.2	4.9	4.7	4.4	6.9		
		MCDBR	15.3	13.4	12.0	11.4	10.9	9.9	8.4	8.0	9.2	10.0	11.9	16.1		
(40) Clear-Sky Solar Irradiance	taub		0.527	0.591	0.725	0.679	0.621	0.582	0.523	0.491	0.503	0.538	0.500	0.483		
	taud		1.912	1.769	1.548	1.697	1.806	1.917	2.105	2.208	2.147	2.019	2.040	2.040		
	Ebn at Noon		778	745	659	687	713	732	781	818	815	779	798	807		
	Edh at Noon		192	229	290	248	218	192	160	147	157	176	167	165		
(44) All-Sky Solar Radiation	RadAvg		5.70	6.06	6.30	6.20	6.08	5.70	5.33	5.10	5.45	5.81	5.68	5.54		
	RadStd		0.28	0.28	0.27	0.31	0.24	0.22	0.20	0.19	0.24	0.15	0.19	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.30	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page